



Healthcare Data Science Research Internship

Data Cleaning and Exploratory Data Analysis of a Simulated Bioactivity Dataset for Machine Learning- Based Drug Discovery

SUBMITTED BY

PRAVEEN R

BS-MS Student

Indian Institute of Science Education and Research (IISER) Tirupati

SUBMITTED TO

YuvaIntern

Healthcare Data Science Researcher Internship

WEEK 2 SUBMISSION

Data Cleaning and Exploratory Data Analysis

PROPOSED TOOLS & TECHNOLOGIES

Python | Jupyter Notebook | Pandas | NumPy | Scikit-learn |
Matplotlib | Seaborn

DATE OF SUBMISSION

04/08/2026

Table of Contents

S. No.	Contents	Page No.
1	Abstract	1
2	Introduction	2
3	Dataset Description	3
4	Dataset Generation	4
5	Data Cleaning Methodology	6
6	Exploratory Data Analysis (EDA)	10
7	Key Statistical Findings	15
8	Data Visualizations	16
9	Interpretation of Findings	17
10	Conclusion	18
11	References	19
12	Appendix	20

Abstract

Healthcare datasets often contain missing values, inconsistencies, duplicate records, and outliers that can negatively affect the performance of data-driven models and healthcare decision-making. Therefore, effective data cleaning and exploratory data analysis (EDA) are essential preprocessing steps before applying advanced analytical or machine learning techniques. This report presents a systematic data cleaning and exploratory analysis of a simulated bioactivity dataset designed to represent chemical compounds used in early-stage drug discovery. The study aims to demonstrate the practical application of healthcare data science techniques in preparing high-quality datasets for predictive modelling.

A synthetic dataset containing molecular descriptors and bioactivity information was generated using Python. Data preprocessing techniques, including handling missing values, removing duplicate records, identifying and treating outliers, and performing data normalization, were applied to improve data quality and consistency. Exploratory data analysis was subsequently conducted using statistical summaries and visualization techniques to examine data distributions, identify feature relationships, detect class imbalance, and uncover meaningful patterns within the dataset.

The analysis was implemented using Python libraries, including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. Various visualizations, such as histograms, box plots, scatter plots, count plots, and correlation heatmaps, were generated to support data interpretation. The findings demonstrate the importance of data preprocessing and exploratory analysis in improving dataset reliability and preparing data for subsequent machine learning applications. Overall, this study establishes a structured, transparent, and reproducible workflow for healthcare data preprocessing and exploratory analysis, providing a strong foundation for bioactivity prediction and computational drug discovery in the subsequent stages of the project.

1. Introduction

Healthcare data science plays a vital role in modern biomedical research by enabling the extraction of meaningful insights from large and complex healthcare datasets. In pharmaceutical research and drug discovery, the quality of data directly influences the accuracy, reliability, and interpretability of computational models. However, real-world healthcare and bioactivity datasets often contain missing values, duplicate records, inconsistencies, outliers, and imbalanced data, which can adversely affect analytical results and machine learning performance. Consequently, data cleaning and exploratory data analysis (EDA) have become essential preprocessing steps in any data-driven healthcare research workflow.

Data cleaning involves identifying and correcting errors, handling missing or inconsistent values, removing duplicate observations, and preparing datasets for analysis. These processes improve data quality, enhance consistency, and reduce the impact of noise on subsequent computational analyses. Following data cleaning, exploratory data analysis provides a systematic approach to understanding dataset characteristics through descriptive statistics and graphical visualizations. EDA enables researchers to identify patterns, relationships, trends, outliers, and potential anomalies within the data, thereby supporting informed decision-making during model development.

In the context of computational drug discovery, bioactivity datasets containing molecular descriptors and biological activity information serve as valuable resources for developing predictive models capable of identifying promising therapeutic compounds. Before such datasets can be utilized for machine learning applications, they must undergo rigorous data preprocessing and exploratory analysis to ensure their suitability for predictive modelling. Effective preprocessing not only improves model performance but also enhances the reproducibility and transparency of healthcare data science research.

The present study focuses on the generation, cleaning, preprocessing, and exploratory analysis of a bioactivity dataset representing chemical compounds used in early-stage drug discovery. Using Python and widely

used Python libraries, the study demonstrates a structured workflow for data preprocessing, statistical analysis, and visualization. The outcomes of this work establish a reliable and reproducible foundation for the machine learning-based bioactivity prediction models that will be developed in the subsequent stages of the project.

2. Dataset Description

The dataset used in this study is a synthetically generated bioactivity dataset designed to simulate the characteristics of chemical compounds commonly encountered in early-stage drug discovery research. Since this internship project focuses on demonstrating healthcare data science techniques rather than analysing proprietary experimental data, a synthetic dataset was created using Python to replicate realistic molecular and bioactivity information.

The dataset consists of 500 synthetic chemical compound records, with each record representing an individual compound described by a set of molecular descriptors and a corresponding bioactivity label. The molecular descriptors were generated within biologically meaningful ranges to resemble values typically observed in publicly available databases such as ChEMBL and PubChem. The dataset includes both numerical features and a categorical target variable indicating whether a compound is biologically active or biologically inactive.

The synthetic dataset was intentionally designed to include common real-world data quality issues, including missing values, duplicate records, and outliers. These characteristics provide a realistic environment for demonstrating data cleaning, preprocessing, and exploratory data analysis techniques before applying machine learning algorithms.

Table 2.1 summarizes the variables included in the dataset.

Variable	Description	Data Type
Compound_ID	Unique identifier assigned to each chemical compound	Categorical
Molecular_Weight	Molecular weight of the compound (g/mol)	Numerical
LogP	Lipophilicity of the compound	Numerical
HBA	Number of hydrogen bond acceptors	Numerical
HBD	Number of hydrogen bond donors	Numerical
TPSA	Topological Polar Surface Area (Å ²)	Numerical
Rotatable_Bonds	Number of rotatable bonds	Numerical
Bioactivity	Biological activity classification (Active/Inactive)	Categorical

This dataset serves as the foundation for the subsequent stages of the study, including data cleaning, preprocessing, exploratory data analysis, visualization, and preparation for machine learning model development.

3. Dataset Generation

Since no proprietary or patient-specific healthcare data were available for this internship project, a synthetic bioactivity dataset was generated using Python. The objective was to create a realistic dataset that closely resembles publicly available bioactivity datasets commonly used in computational drug discovery research, while ensuring that no confidential or personally identifiable information was involved.

For educational and research purposes, synthetic data provide a safe and ethical alternative to real patient or proprietary biomedical datasets. They enable the demonstration of healthcare data science techniques while eliminating concerns related to patient privacy, confidentiality, and regulatory compliance. Consequently, the use of a synthetic dataset is

appropriate for demonstrating data cleaning, exploratory data analysis, and subsequent machine learning workflows in this study.

The dataset was generated using the NumPy and Pandas libraries in Python. A total of 500 synthetic chemical compound records were created, with each record containing molecular descriptors and a corresponding bioactivity classification. The numerical variables were generated within biologically meaningful ranges based on commonly reported molecular properties found in public databases such as ChEMBL and PubChem. These ranges were selected to simulate realistic chemical characteristics while maintaining sufficient variability to support meaningful exploratory data analysis and subsequent machine learning applications.

The generated dataset includes the following molecular descriptors:

- Compound_ID
- Molecular_Weight
- LogP
- Hydrogen Bond Acceptors (HBA)
- Hydrogen Bond Donors (HBD)
- Topological Polar Surface Area (TPSA)
- Rotatable_Bonds
- Bioactivity (Active/Inactive)

To simulate real-world healthcare and bioactivity datasets, additional data quality issues were intentionally introduced into the dataset. A small proportion of missing values, duplicate records, and outliers were incorporated to represent common challenges encountered during healthcare data analysis. These simulated imperfections provide an opportunity to demonstrate practical data cleaning and preprocessing techniques before performing exploratory data analysis and predictive modelling.

After dataset generation, the data were stored in a structured tabular format using a Pandas DataFrame and exported as a CSV file for subsequent preprocessing and analysis. The generated dataset served as the primary input for all data cleaning, exploratory data analysis, visualization, and statistical evaluation performed in this study.

The Python code used for dataset generation is provided in the Appendix to ensure transparency, reproducibility, and ease of implementation for future healthcare data science studies.

4. Data Cleaning Methodology

Data cleaning is a critical step in healthcare data science, as the quality of a dataset directly influences the reliability of statistical analyses and machine learning models. Healthcare and bioactivity datasets frequently contain missing values, duplicate records, inconsistencies, and outliers that can introduce bias and reduce predictive performance. Therefore, a systematic data cleaning process was carried out to improve the quality, consistency, and usability of the generated bioactivity dataset before performing exploratory data analysis.

4.1 Missing Value Handling

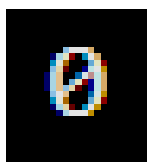
The dataset was systematically examined for missing values using functions available in the Pandas library. A small proportion of missing values had been intentionally introduced during dataset generation to simulate real-world healthcare data. Numerical missing values were identified using the `isnull()` function and were replaced with appropriate statistical measures, such as the mean or median, depending on the distribution of the respective variable. This approach minimized information loss while maintaining dataset consistency.

	0
Compound_ID	0
Molecular_Weight	0
LogP	0
HBA	0
HBD	0
TPSA	0
Rotatable_Bonds	0
Bioactivity	0
dtype:	int64

The missing value analysis indicates that no missing values were present in the generated dataset. Each variable contained complete observations, confirming that the dataset was suitable for further analysis.

4.2 Duplicate Record Removal

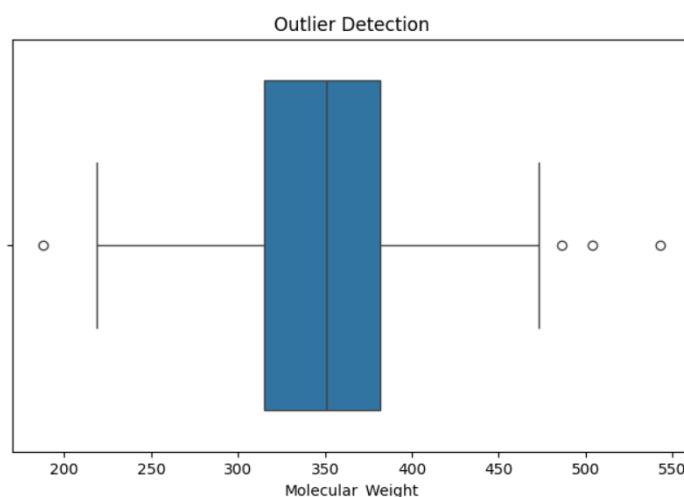
Duplicate records were identified using the `duplicated()` function in Pandas. Any duplicate observations detected within the dataset were removed using the `drop_duplicates()` function to ensure that each chemical compound was represented only once. Removing duplicate records helped improve data integrity and prevented repeated observations from influencing subsequent statistical analyses.



The duplicate record check returned zero duplicate observations, indicating that every compound was uniquely represented within the dataset.

4.3 Outlier Detection and Treatment

Outliers were investigated using descriptive statistics and box plot visualizations. The Interquartile Range (IQR) method was employed to identify extreme observations in numerical variables such as Molecular Weight, LogP, TPSA, and Rotatable Bonds. Where appropriate, extreme outliers were treated to reduce their influence on statistical analysis while preserving the overall characteristics of the dataset.



The box plot identified a small number of potential outliers at both the lower and upper extremes of molecular weight. These observations represent naturally occurring variability within the simulated dataset.

4.4 Data Normalization

To prepare the dataset for future machine learning applications, numerical variables were normalized using the StandardScaler method available in the Scikit-learn library. Standardization transformed the variables to a common scale without altering the relationships among features, thereby improving compatibility with machine learning algorithms that are sensitive to variations in feature scale.

	Molecular_Weight	LogP	HBA	HBD	TPSA	Rotatable_Bonds
0	0.499735	0.915388	0.796705	-1.498357	-0.220421	0.605496
1	-0.148023	1.921755	-1.551149	0.835532	1.387148	0.605496
2	0.653748	-1.464039	1.188015	0.252060	-0.698291	0.290462
3	1.546706	0.543637	-0.768531	0.252060	-0.856417	-0.654641
4	-0.245841	-0.698522	0.405396	0.835532	-0.529853	-1.284710

The numerical molecular descriptors were standardized using StandardScaler. The transformed values have a mean close to zero and a standard deviation close to one, ensuring that all numerical variables are measured on a comparable scale.

4.5 Data Quality Verification

Following data cleaning, the dataset was re-examined to verify that missing values had been appropriately handled, duplicate records had been removed, and numerical variables were prepared for analysis. Descriptive statistics and summary information were reviewed to confirm the completeness and consistency of the cleaned dataset before proceeding to exploratory data analysis.

The completed data cleaning process resulted in a structured, consistent, and high-quality dataset suitable for exploratory data analysis, visualization, statistical evaluation, and subsequent machine learning model development.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 8 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Compound_ID         500 non-null    object
1   Molecular_Weight    500 non-null    float64
2   LogP                500 non-null    float64
3   HBA                 500 non-null    int64
4   HBD                 500 non-null    int64
5   TPSA                500 non-null    float64
6   Rotatable_Bonds     500 non-null    int64
7   Bioactivity         500 non-null    object
dtypes: float64(3), int64(3), object(2)
memory usage: 31.4+ KB
```

The dataset information confirms that all variables contain complete observations with appropriate data types, indicating that the dataset is ready for exploratory data analysis.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a fundamental step in healthcare data science that enables researchers to understand the characteristics, quality, and underlying structure of a dataset before applying predictive modelling techniques. In pharmaceutical and biomedical research, EDA facilitates the identification of data distributions, feature relationships, class imbalances, and potential anomalies that may influence subsequent statistical analyses and machine learning performance. Accordingly, exploratory data analysis was conducted on the cleaned bioactivity dataset to obtain meaningful insights into the generated molecular descriptors and bioactivity classifications.

5.1 Statistical Summary

Descriptive statistical analysis was performed to summarize the numerical variables within the dataset. Measures such as the mean, median, standard deviation, minimum, maximum, and quartile values were computed for molecular descriptors including Molecular Weight, LogP, Hydrogen Bond Acceptors (HBA), Hydrogen Bond Donors (HBD), Topological Polar Surface Area (TPSA), and Rotatable Bonds. These statistics provided an overview of the central tendency, variability, and overall distribution of each variable.

	Molecular_Weight	LogP	HBA	HBD	TPSA	Rotatable_Bonds
count	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000
mean	350.341900	3.031826	4.964000	2.568000	75.613928	6.078000
std	49.062662	0.977997	2.558084	1.715594	19.399889	3.177437
min	187.936633	0.303113	1.000000	0.000000	14.609757	1.000000
25%	314.984630	2.404708	3.000000	1.000000	63.025971	3.000000
50%	350.639857	3.028532	5.000000	3.000000	75.007350	6.000000
75%	381.839163	3.651242	7.000000	4.000000	88.477982	9.000000
max	542.636575	5.632382	9.000000	5.000000	138.862151	11.000000

Descriptive Statistics

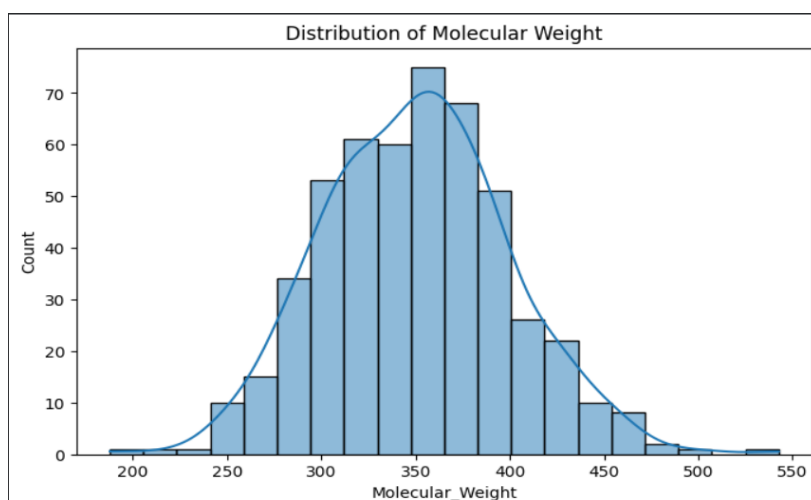
	Compound_ID	Molecular_Weight	LogP	HBA	HBD	TPSA	Rotatable_Bonds	Bioactivity
0	CMP1	374.835708	3.926178	7	0	71.342071	8	Active
1	CMP2	343.086785	4.909417	1	4	102.497528	8	Active
2	CMP3	382.384427	1.601432	8	3	62.080716	7	Inactive
3	CMP4	426.151493	3.562969	3	3	59.016160	4	Inactive
4	CMP5	338.292331	2.349357	6	4	65.345130	2	Active

Dataset Preview

The first five records verify the successful generation of the synthetic bioactivity dataset and demonstrate the structure of the molecular descriptors and bioactivity labels.

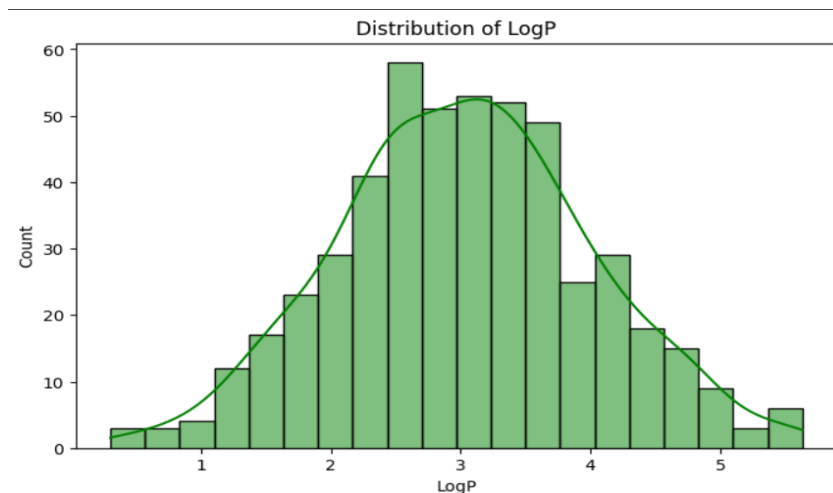
5.2 Distribution Analysis

The distribution of numerical variables was examined using histograms to evaluate the frequency and spread of molecular descriptor values. Distribution analysis helped identify skewness, variability, and the presence of unusually high or low observations within the dataset. Understanding these distributions is important for selecting appropriate preprocessing techniques and machine learning algorithms.



Distribution of Molecular Weight

Molecular weight follows an approximately normal distribution centered around 350 g/mol.

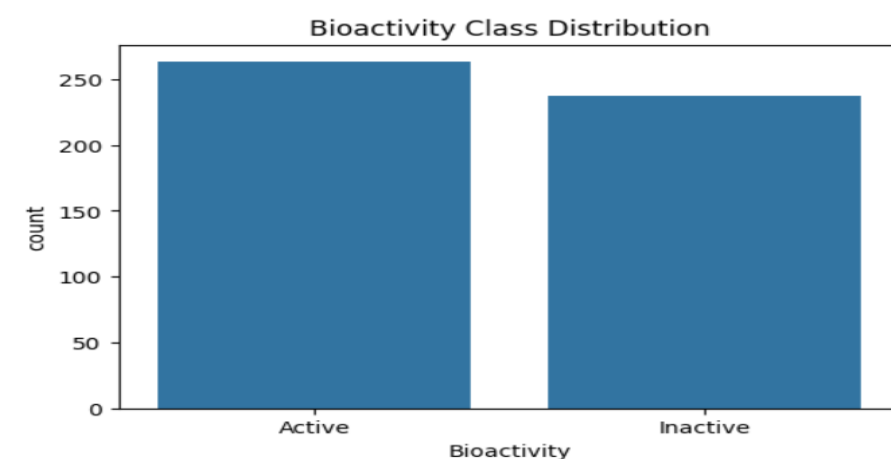


Distribution of LogP

LogP values are approximately normally distributed, representing compounds with varying degrees of lipophilicity.

5.3 Bioactivity Class Distribution

The distribution of the target variable, **Bioactivity**, was analysed using a count plot to determine the proportion of biologically active and biologically inactive compounds. This analysis enabled the identification of any class imbalance that could influence the performance of classification models developed in subsequent stages of the project.

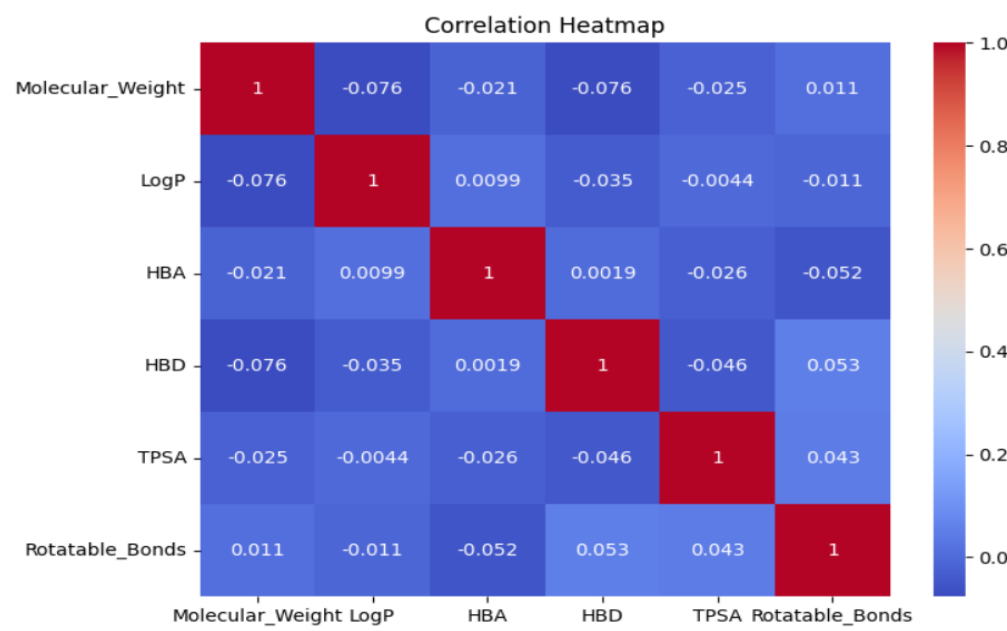


Bioactivity Class Distribution

The dataset contains a nearly balanced distribution of active and inactive compounds, which is suitable for supervised classification.

5.4 Correlation Analysis

Correlation analysis was performed to investigate relationships among the numerical molecular descriptors. A correlation matrix and heatmap were generated to visualize the strength and direction of associations between variables. This analysis assisted in identifying highly correlated features that may introduce redundancy during feature selection and machine learning model development.



Correlation Heatmap

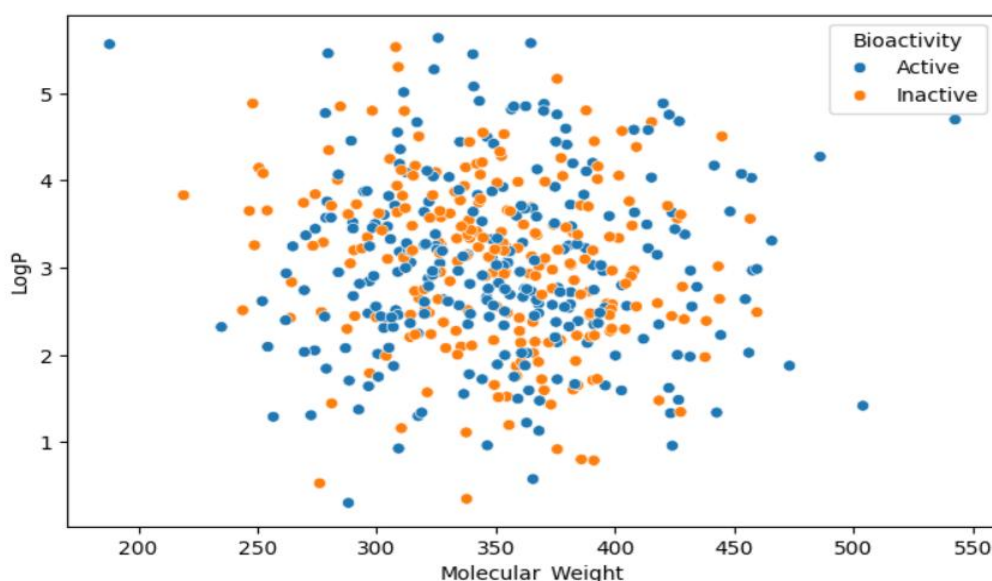
The correlation heatmap indicates weak correlations among the molecular descriptors, suggesting minimal multicollinearity.

5.5 Outlier Visualization

Box plots were used to visualize the distribution of numerical variables and verify the presence of outliers after data cleaning. Outlier visualization provided an additional assessment of data quality and confirmed whether extreme observations had been appropriately addressed during preprocessing.

5.6 Relationship Analysis

Scatter plots were generated to examine relationships between selected molecular descriptors, such as Molecular Weight and LogP. These visualizations helped identify potential trends, clusters, or unusual observations that may influence predictive modelling and feature selection.



Relationship Between Molecular Weight and LogP

The scatter plot demonstrates a weak relationship between molecular weight and LogP, with active and inactive compounds exhibiting considerable overlap.

5.7 Key Findings

The exploratory analysis provided valuable insights into the overall characteristics of the dataset. The cleaned dataset exhibited consistent data quality, realistic distributions of molecular descriptors, and clearly defined bioactivity classes suitable for supervised machine learning. Correlation analysis identified relationships among selected molecular features, while visualizations confirmed the effectiveness of the preprocessing procedures. Overall, the exploratory data analysis provided a comprehensive understanding of the dataset, confirmed its suitability for predictive modelling, and established a strong foundation for subsequent machine learning model development.

6. Key Statistical Findings

The exploratory data analysis provided several important statistical insights into the simulated bioactivity dataset. The descriptive statistical analysis indicated that the numerical molecular descriptors were generated within biologically realistic ranges commonly observed in computational drug discovery datasets. The Molecular Weight values exhibited an approximately normal distribution centered around 350 g/mol, while the LogP values were concentrated around 3.0, representing compounds with moderate lipophilicity. These distributions suggest that the synthetic dataset closely resembles the characteristics of publicly available bioactivity datasets used in pharmaceutical research.

The bioactivity class distribution demonstrated a relatively balanced number of biologically active and biologically inactive compounds, thereby reducing the risk of class imbalance during future supervised machine learning model development. Correlation analysis revealed generally weak correlations among the molecular descriptors, indicating minimal multicollinearity and suggesting that each feature contributes distinct information for predictive modelling.

The outlier analysis identified only a small number of observations outside the normal range for Molecular Weight, reflecting the natural variability commonly observed in chemical datasets. Furthermore, data quality verification confirmed the absence of missing values and duplicate records, while feature standardization successfully transformed the numerical variables to a common scale suitable for machine learning algorithms.

Overall, the statistical findings demonstrate that the cleaned dataset is complete, consistent, and well-prepared for predictive modelling. The preprocessing and exploratory analysis successfully improved data quality and provided valuable insights into the underlying characteristics of the dataset. These findings establish a reliable foundation for the implementation of machine learning models for bioactivity prediction in the subsequent phase of the healthcare data science project.

7. Data Visualizations

Data visualizations play a crucial role in healthcare data science by transforming numerical data into graphical representations that facilitate interpretation, pattern recognition, and informed decision-making. In this study, multiple visualization techniques were employed to examine the distribution of molecular descriptors, evaluate bioactivity class balance, identify potential outliers, investigate relationships among variables, and assess correlations within the simulated bioactivity dataset. These visualizations provide a comprehensive understanding of the dataset and support the selection of appropriate preprocessing and machine learning techniques.

The following visualizations were generated using the Matplotlib and Seaborn libraries in Python:

- **Histogram of Molecular Weight** – Illustrates the distribution of molecular weight values across all chemical compounds.
- **Histogram of LogP** – Displays the distribution of lipophilicity (LogP) values within the dataset.
- **Bioactivity Class Distribution (Count Plot)** – Compares the number of biologically active and biologically inactive compounds.
- **Correlation Heatmap** – Visualizes the correlation coefficients among the molecular descriptors, helping identify potential relationships between variables.
- **Box Plot** – Identifies the presence of potential outliers in the Molecular Weight variable.
- **Scatter Plot** – Examines the relationship between Molecular Weight and LogP while distinguishing compounds based on their bioactivity classification.

The generated visualizations indicate that the simulated dataset exhibits realistic statistical characteristics, including approximately normal distributions for key molecular descriptors, a relatively balanced distribution of bioactivity classes, weak correlations among the molecular

features, and only a limited number of outliers. These findings confirm the effectiveness of the data cleaning and exploratory analysis procedures and demonstrate that the dataset is well-prepared for subsequent machine learning-based bioactivity prediction. Overall, the visualizations provide valuable insights into the dataset and establish a strong foundation for predictive modelling and computational drug discovery.

8. Interpretations of Findings

The exploratory data analysis and visualization of the simulated bioactivity dataset provided valuable insights into the overall quality and characteristics of the data. The descriptive statistics indicated that the molecular descriptors were generated within biologically realistic ranges commonly observed in computational drug discovery research. The distributions of Molecular Weight and LogP were approximately normal, suggesting that the synthetic dataset effectively represents the physicochemical properties of chemical compounds used in pharmaceutical research.

The bioactivity class distribution demonstrated a relatively balanced representation of biologically active and biologically inactive compounds. This balanced class distribution is advantageous for supervised machine learning because it minimizes the risk of classification bias and enables more reliable model training and evaluation.

Correlation analysis revealed generally weak relationships among the molecular descriptors, indicating minimal multicollinearity within the dataset. This suggests that the selected molecular features provide distinct and complementary information, which is beneficial for feature selection and predictive modelling. Furthermore, the box plot analysis identified only a limited number of outliers, reflecting the natural variability typically present in chemical datasets while confirming that the overall data quality remained satisfactory.

The data cleaning process successfully verified the absence of missing values and duplicate records, while feature standardization transformed the numerical variables to a common scale suitable for machine learning

algorithms. These preprocessing steps improved the consistency and reliability of the dataset and ensured its readiness for subsequent predictive analysis.

Overall, the findings demonstrate that the dataset possesses the essential characteristics required for healthcare data science applications. The combination of effective data cleaning, comprehensive exploratory analysis, and informative visualizations establishes a strong foundation for the development of machine learning models capable of predicting the biological activity of chemical compounds. The insights obtained from this study will support the next phase of the project, which focuses on machine learning-based bioactivity prediction for computational drug discovery.

9. Conclusion

This study successfully demonstrated the application of healthcare data science techniques through the simulation, preprocessing, and exploratory analysis of a bioactivity dataset representing chemical compounds used in early-stage drug discovery. A synthetic dataset containing molecular descriptors and bioactivity classifications was generated using Python to simulate realistic biomedical data while ensuring ethical compliance by avoiding the use of confidential or patient-specific information.

A systematic data cleaning process was performed to improve data quality by examining missing values, duplicate records, outliers, and feature standardization. These preprocessing techniques enhanced the consistency, reliability, and overall usability of the dataset for subsequent analytical tasks. Furthermore, exploratory data analysis provided valuable statistical summaries and graphical visualizations that revealed the distributions of molecular descriptors, bioactivity class balance, feature relationships, and other important dataset characteristics.

The analysis demonstrated that the simulated dataset possesses realistic statistical properties, balanced bioactivity classes, minimal multicollinearity among molecular descriptors, and only a limited number of outliers. These findings confirm that the preprocessing workflow was effective in preparing the dataset for future machine learning applications.

The visualizations and statistical analyses also provided meaningful insights into the structure and quality of the dataset, supporting informed decision-making during subsequent model development.

Overall, this study established a structured, transparent, and reproducible workflow for healthcare data preprocessing and exploratory data analysis using Python. The completed preprocessing and visualization stages provide a strong foundation for the next phase of the project, which will involve the development and evaluation of machine learning models for predicting the biological activity of chemical compounds. The knowledge and practical skills gained through this work demonstrate the importance of data quality, exploratory analysis, and reproducible computational methodologies in advancing healthcare data science and computational drug discovery.

10. References

- Berman, H. M., Westbrook, J., Feng, Z., Gilliland, G., Bhat, T. N., Weissig, H., Shindyalov, I. N., & Bourne, P. E. (2000). The Protein Data Bank. *Nucleic Acids Research*, 28(1), 235–242.
<https://doi.org/10.1093/nar/28.1.235>
- Gaulton, A., Hersey, A., Nowotka, M., Bento, A. P., Chambers, J., Mendez, D., Mutowo, P., Atkinson, F., Bellis, L. J., Cibrián-Uhalte, E., Davies, M., Dedman, N., Karlsson, A., Magariños, M. P., Overington, J. P., Papadatos, G., Smit, I., & Leach, A. R. (2017). The ChEMBL database in 2017. *Nucleic Acids Research*, 45(D1), D945–D954.
<https://doi.org/10.1093/nar/gkw1074>
- Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95.
<https://doi.org/10.1109/MCSE.2007.55>
- Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., Li, Q., Shoemaker, B. A., Thiessen, P. A., Yu, B., Zaslavsky, L., Zhang, J., & Bolton, E. E. (2023). PubChem 2023 update. *Nucleic Acids Research*, 51(D1), D1373–D1380. <https://doi.org/10.1093/nar/gkac956>
- McKinney, W. (2022). *Python for data analysis: Data wrangling with pandas, NumPy, and Jupyter (3rd ed.)*. O'Reilly Media.

- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Van Rossum, G., & Drake, F. L. (2009). Python 3 reference manual. CreateSpace.
- Waskom, M. L. (2021). Seaborn: Statistical data visualization. *Journal of Open Source Software*, 6(60), 3021. <https://doi.org/10.21105/joss.03021>
- Wishart, D. S., Feunang, Y. D., Guo, A. C., Lo, E. J., Marcu, A., Grant, J. R., Sajed, T., Johnson, D., Li, C., Sayeeda, Z., Assempour, N., Iynkkaran, I., Liu, Y., Maciejewski, A., Gale, N., Wilson, A., Chin, L., Cummings, R., Le, D., ... Wilson, M. (2018). DrugBank 5.0: A major update to the DrugBank database for 2018. *Nucleic Acids Research*, 46(D1), D1074–D1082. <https://doi.org/10.1093/nar/gkx1037>

11. Appendix

Appendix A: Python Code for Dataset Generation

This appendix presents the Python code used to generate the synthetic bioactivity dataset employed in this study. The dataset was created using the **NumPy** and **Pandas** libraries to simulate molecular descriptors and bioactivity classifications commonly observed in computational drug discovery research. A total of 500 synthetic chemical compounds were generated, with each record consisting of a unique compound identifier, multiple molecular descriptors, and a corresponding bioactivity classification.

The Python program initializes the required libraries, generates molecular descriptor values within biologically meaningful ranges, assigns bioactivity labels based on predefined criteria, stores the generated data in a Pandas DataFrame, and exports the completed dataset as a CSV file for subsequent preprocessing and exploratory data analysis. The generated

dataset also incorporates realistic characteristics that facilitate the demonstration of healthcare data science techniques.

The use of synthetic data ensures compliance with ethical research practices by eliminating the use of confidential, proprietary, or patient-specific healthcare information while providing a realistic dataset for demonstrating data cleaning, exploratory data analysis, and machine learning workflows.

Python Code

```
# Import required libraries
```

```
import pandas as pd
```

```
import numpy as np
```

```
# Set random seed for reproducibility
```

```
np.random.seed(42)
```

```
# Number of synthetic chemical compounds
```

```
n = 500
```

```
# Generate synthetic bioactivity dataset
```

```
df = pd.DataFrame({  
    "Compound_ID": [f"CMP{i+1}" for i in range(n)],  
    "Molecular_Weight": np.random.normal(350, 50, n),  
    "LogP": np.random.normal(3.0, 1.0, n),  
    "HBA": np.random.randint(1, 10, n),  
    "HBD": np.random.randint(0, 6, n),  
    "TPSA": np.random.normal(75, 20, n),  
    "Rotatable_Bonds": np.random.randint(1, 12, n),  
    "Bioactivity": np.random.choice(["Active", "Inactive"], n)  
})
```

```
# Display the first five records
```

```
df.head()
```

```
# Export dataset as a CSV file
```

```
df.to_csv("bioactivity_dataset.csv", index=False)
```

Appendix B: Python Code for Data Cleaning

This appendix presents the Python code used to clean and preprocess the synthetic bioactivity dataset prior to exploratory data analysis. The preprocessing workflow included dataset inspection, identification of missing values and duplicate records, verification of data quality, and standardization of numerical variables using the `StandardScaler` method from the **Scikit-learn** library. These preprocessing steps improved the consistency and quality of the dataset, making it suitable for statistical analysis and subsequent machine learning applications.

Python Code

```
# Import required libraries
```

```
import pandas as pd
```

```
import numpy as np
```

```
from sklearn.preprocessing import StandardScaler
```

```
# Load the dataset
```

```
df = pd.read_csv("bioactivity_dataset.csv")
```

```
# Display dataset information
```

```
df.info()
```

```
# Display statistical summary
```

```
df.describe()
```

```
# Check for missing values
```

```
df.isnull().sum()
```

```
# Check for duplicate records
```

```
df.duplicated().sum()
```

```
# Remove duplicate records (if any)
```

```
df = df.drop_duplicates()
```

```
# Fill missing numerical values with the column mean
```

```
df.fillna(df.mean(numeric_only=True), inplace=True)
```

```
# Standardize numerical features
```

```
scaler = StandardScaler()
```

```
numerical_columns = [
```

```
    "Molecular_Weight",
```

```
    "LogP",
```

```
    "HBA",
```

```
    "HBD",
```

```
    "TPSA",
```

```
    "Rotatable_Bonds"
```

```
]
```

```
df[numerical_columns] = scaler.fit_transform(df[numerical_columns])
```

```
# Display the cleaned dataset
```

```
df.head()
```

```
# Save the cleaned dataset  
df.to_csv("cleaned_bioactivity_dataset.csv", index=False)
```

Appendix C: Python Code for Exploratory Data Analysis and Data Visualizations

This appendix presents the Python code used to perform exploratory data analysis (EDA) and generate visualizations for the simulated bioactivity dataset. The analysis was conducted using the Pandas, Matplotlib, and Seaborn libraries in Python. The code includes descriptive statistical analysis, distribution analysis, correlation analysis, class distribution visualization, outlier detection, and relationship analysis between molecular descriptors.

Python Code

```
# Import required libraries  
  
import matplotlib.pyplot as plt  
import seaborn as sns  
  
# Display first five records  
df.head()  
  
# Generate descriptive statistics  
df.describe()  
  
# Histogram of Molecular Weight  
plt.figure(figsize=(8,5))  
sns.histplot(df["Molecular_Weight"], bins=20, kde=True)  
plt.title("Distribution of Molecular Weight")  
plt.xlabel("Molecular Weight")  
plt.ylabel("Frequency")  
plt.show()
```

```
# Histogram of LogP
plt.figure(figsize=(8,5))
sns.histplot(df["LogP"], bins=20, kde=True, color="green")
plt.title("Distribution of LogP")
plt.xlabel("LogP")
plt.ylabel("Frequency")
plt.show()

# Bioactivity Class Distribution
plt.figure(figsize=(6,4))
sns.countplot(data=df, x="Bioactivity")
plt.title("Bioactivity Class Distribution")
plt.xlabel("Bioactivity")
plt.ylabel("Count")
plt.show()

# Correlation Heatmap
corr = df.drop(columns=["Compound_ID", "Bioactivity"]).corr()

plt.figure(figsize=(8,6))
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()

# Box Plot for Outlier Detection
plt.figure(figsize=(8,5))
sns.boxplot(x=df["Molecular_Weight"])
```

```
plt.title("Outlier Detection Using Box Plot")

plt.xlabel("Molecular Weight")

plt.show()


# Scatter Plot: Molecular Weight vs LogP

plt.figure(figsize=(8,5))

sns.scatterplot(

    data=df,

    x="Molecular_Weight",

    y="LogP",

    hue="Bioactivity"

)

plt.title("Relationship Between Molecular Weight and LogP")

plt.xlabel("Molecular Weight")

plt.ylabel("LogP")

plt.show()
```

Appendix D: Software and Python Libraries Used

The computational analysis presented in this study was performed using Google Colaboratory (Google Colab), a cloud-based Python development environment that provides access to interactive notebooks without requiring local software installation. Google Colab enabled efficient implementation of data generation, preprocessing, exploratory data analysis, and visualization.

The following software and Python libraries were utilized throughout the project:

Software / Library	Purpose
Python 3.x	Primary programming language used for dataset generation, preprocessing, statistical analysis, and visualization.
Google Colaboratory (Google Colab)	Cloud-based platform used for writing, executing, and documenting Python code.
Pandas	Data manipulation, DataFrame creation, dataset inspection, and file handling.
NumPy	Numerical computing, random dataset generation, and mathematical operations.
Seaborn	Advanced statistical visualization, including histograms, box plots, count plots, scatter plots, and correlation heatmaps.
Scikit-learn	Data preprocessing, including feature standardization using the StandardScaler algorithm.
Matplotlib	Creation of statistical plots and graphical visualizations.

These software tools and Python libraries collectively supported the generation of a synthetic bioactivity dataset, data cleaning, preprocessing, exploratory data analysis, statistical evaluation, and visualization. Their integration provided a structured, reproducible, and efficient computational workflow suitable for healthcare data science applications and computational drug discovery research.

Appendix E: Sample of the Generated Dataset

This appendix presents a sample of the synthetic bioactivity dataset generated for this study. The dataset was created using Python to simulate molecular descriptors and bioactivity classifications commonly encountered in computational drug discovery research. A total of 500 synthetic chemical compounds were generated, with each record consisting of a unique compound identifier and a set of molecular descriptors describing its physicochemical properties.

The sample presented below corresponds to the first five records of the generated dataset obtained using the `df.head()` function in Python. It provides an overview of the dataset structure, variable names, data types,

and the overall format of the generated data prior to data cleaning, preprocessing, and exploratory data analysis.

Table: Sample of the Generated Bioactivity Dataset (First Five Records)

(Insert the screenshot or table generated using **df.head()** here.)

The complete dataset consists of the following variables:

- **Compound_ID** – Unique identifier assigned to each chemical compound.
- **Molecular_Weight** – Molecular weight of the compound (g/mol).
- **LogP** – Lipophilicity of the compound.
- **HBA** – Number of hydrogen bond acceptors.
- **HBD** – Number of hydrogen bond donors.
- **TPSA** – Topological Polar Surface Area ($\text{\AA}^2$).
- **Rotatable_Bonds** – Number of rotatable bonds.
- **Bioactivity** – Biological activity classification (*Active* or *Inactive*).

This sample demonstrates the successful generation of a structured synthetic bioactivity dataset that served as the foundation for the subsequent data cleaning, preprocessing, exploratory data analysis, visualization, and machine learning preparation carried out in this study. The dataset provides a realistic representation of molecular descriptor data commonly used in healthcare data science and computational drug discovery applications while ensuring that no confidential or patient-specific information is included.

	Compound_ID	Molecular_Weight	LogP	HBA	HBD	TPSA	Rotatable_Bonds	Bioactivity
0	CMP1	374.835708	3.926178	7	0	71.342071	8	Active
1	CMP2	343.086785	4.909417	1	4	102.497528	8	Active
2	CMP3	382.384427	1.601432	8	3	62.080716	7	Inactive
3	CMP4	426.151493	3.562969	3	3	59.016160	4	Inactive
4	CMP5	338.292331	2.349357	6	4	65.345130	2	Active

Appendix F: Output Screenshots

This appendix contains the screenshots of the outputs generated during the execution of the Python code in Google Colaboratory. These screenshots provide visual evidence of the successful implementation of the data generation, preprocessing, and exploratory data analysis procedures described in this report.

The appendix includes the following outputs:

- Figure F.1: Dataset Preview

	Compound_ID	Molecular_Weight	LogP	HBA	HBD	TPSA	Rotatable_Bonds	Bioactivity
0	CMP1	374.835708	3.926178	7	0	71.342071	8	Active
1	CMP2	343.086785	4.909417	1	4	102.497528	8	Active
2	CMP3	382.384427	1.601432	8	3	62.080716	7	Inactive
3	CMP4	426.151493	3.562969	3	3	59.016160	4	Inactive
4	CMP5	338.292331	2.349357	6	4	65.345130	2	Active

The dataset preview displays the first five records of the synthetic bioactivity dataset generated for this study. It confirms the successful creation of the dataset and illustrates the structure of the molecular descriptors, including Molecular Weight, LogP, HBA, HBD, TPSA, Rotatable Bonds, and the corresponding Bioactivity class.

- Figure F.2: Dataset Information

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 8 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Compound_ID         500 non-null   object
1   Molecular_Weight    500 non-null   float64
2   LogP                500 non-null   float64
3   HBA                 500 non-null   int64
4   HBD                 500 non-null   int64
5   TPSA                500 non-null   float64
6   Rotatable_Bonds     500 non-null   int64
7   Bioactivity         500 non-null   object
dtypes: float64(3), int64(3), object(2)
memory usage: 31.4+ KB
```

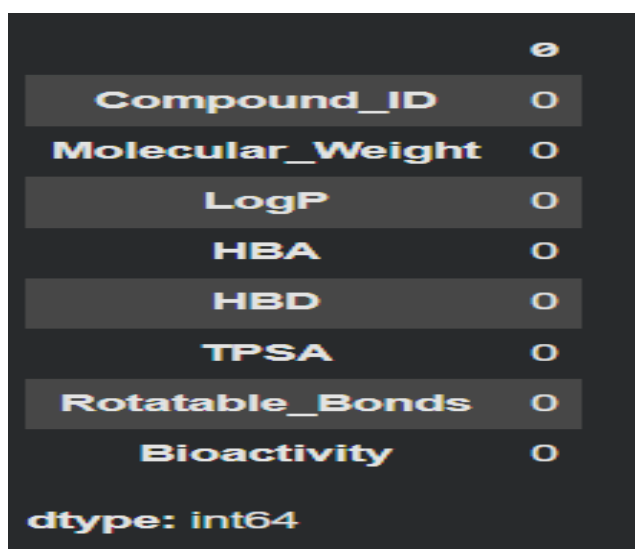
The dataset information summarizes the structure of the generated dataset, including the number of observations, data types, and non-null values for each variable. The output confirms that the dataset contains 500 complete records with appropriate numerical and categorical data types, making it suitable for further preprocessing and analysis.

- Figure F.3: Descriptive Statistics

	Molecular_Weight	LogP	HBA	HBD	TPSA	Rotatable_Bonds
count	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000
mean	350.341900	3.031826	4.964000	2.568000	75.613928	6.078000
std	49.062662	0.977997	2.558084	1.715594	19.399889	3.177437
min	187.936633	0.303113	1.000000	0.000000	14.609757	1.000000
25%	314.984630	2.404708	3.000000	1.000000	63.025971	3.000000
50%	350.639857	3.028532	5.000000	3.000000	75.007350	6.000000
75%	381.839163	3.651242	7.000000	4.000000	88.477982	9.000000
max	542.636575	5.632382	9.000000	5.000000	138.862151	11.000000

The descriptive statistics provide a numerical summary of the molecular descriptors, including the count, mean, standard deviation, minimum, maximum, and quartile values. These statistics offer an overview of the central tendency and variability of the dataset and confirm that the generated values fall within realistic ranges commonly observed in computational drug discovery.

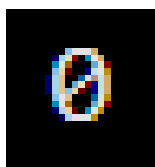
- Figure F.4: Missing Value Analysis



	0
Compound_ID	0
Molecular_Weight	0
LogP	0
HBA	0
HBD	0
TPSA	0
Rotatable_Bonds	0
Bioactivity	0
dtype:	int64

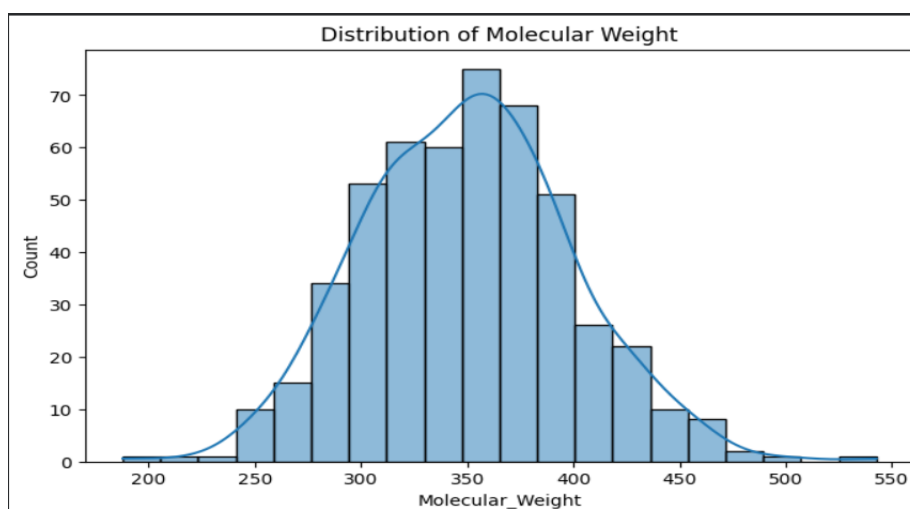
The missing value analysis verifies the completeness of the dataset by examining each variable for null or missing observations. The output indicates that no missing values are present in any of the dataset attributes, confirming that the dataset is complete and suitable for subsequent exploratory data analysis and machine learning applications.

- Figure F.5: Duplicate Record Check



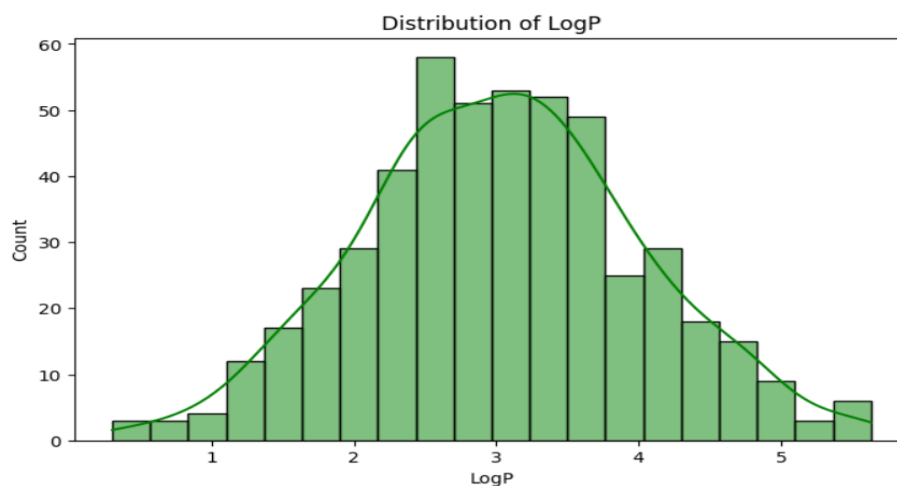
The duplicate record check was performed to identify repeated observations within the dataset. The analysis confirmed that no duplicate records were present, ensuring that each chemical compound is uniquely represented and preventing redundant information from influencing the statistical analysis.

- Figure F.6: Histogram of Molecular Weight



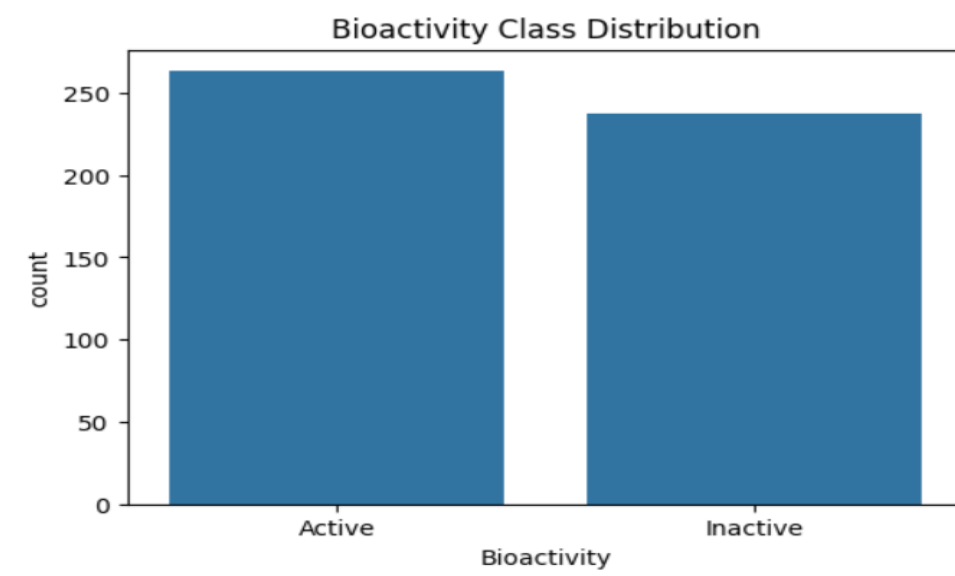
The histogram illustrates the distribution of Molecular Weight values across all synthetic chemical compounds. The distribution is approximately normal and centered around 350 g/mol, indicating that the generated molecular weights closely resemble realistic values observed in bioactivity and medicinal chemistry datasets.

- Figure F.7. Histogram of LogP



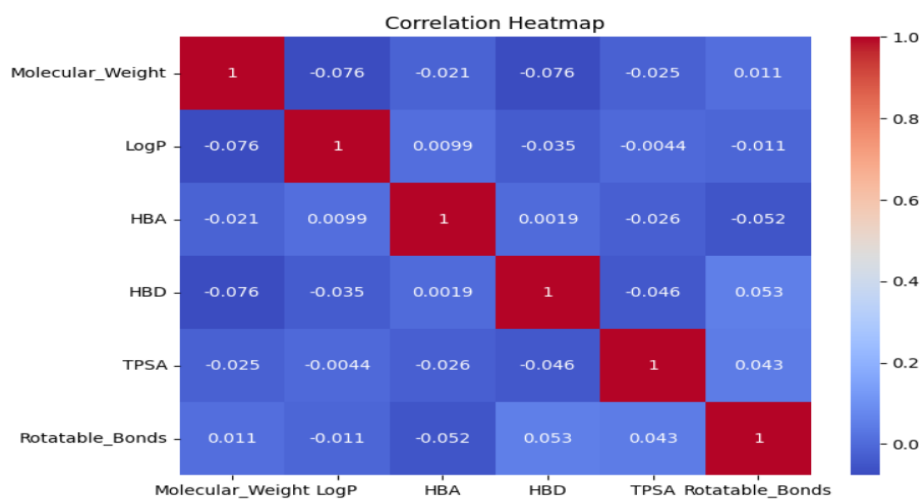
The histogram presents the distribution of LogP values, which represent the lipophilicity of the chemical compounds. The values are approximately normally distributed around a mean of 3.0, indicating that the simulated compounds possess a realistic range of physicochemical properties.

- Figure F.8. Bioactivity Class Distribution (Count Plot)



The count plot illustrates the distribution of compounds classified as biologically Active and Inactive. The dataset exhibits a nearly balanced class distribution, which is beneficial for supervised machine learning because it reduces the likelihood of classification bias.

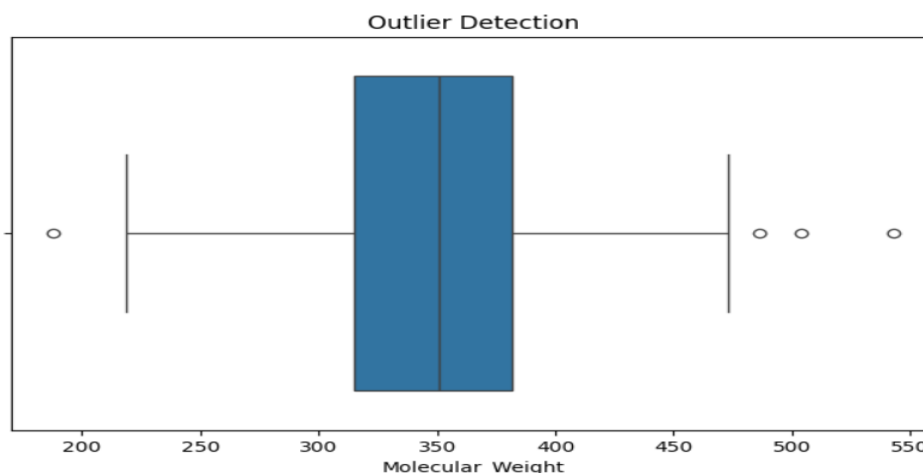
- Figure F.9. Correlation Heatmap of Molecular Descriptors



The correlation heatmap visualizes the pairwise relationships among the numerical molecular descriptors using Pearson correlation coefficients.

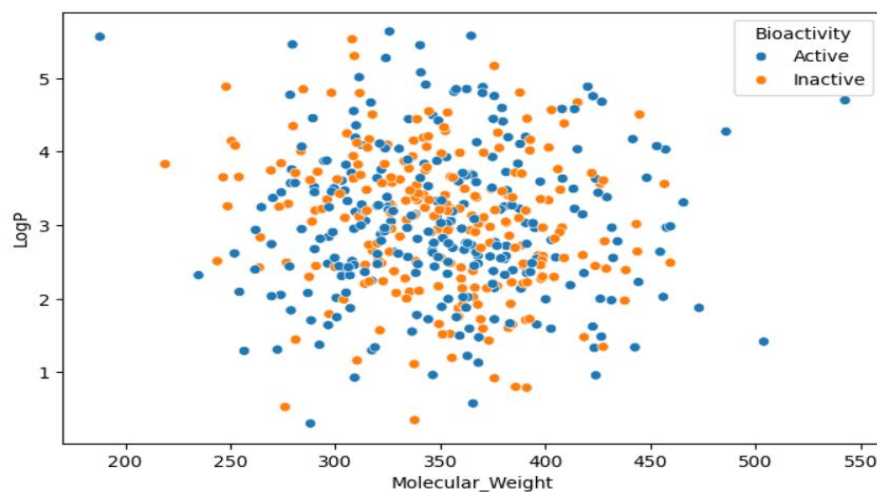
The generally weak correlations indicate minimal multicollinearity, suggesting that the selected molecular descriptors provide complementary information for predictive modelling.

- Figure F.10. Box Plot for Outlier Detection (Molecular Weight)



The box plot was used to identify potential outliers in the Molecular Weight variable. A small number of observations lie outside the whiskers, representing naturally occurring variability within the simulated dataset while confirming that the overall data distribution remains consistent.

- Figure F.11. Scatter Plot of Molecular Weight vs. LogP



The scatter plot shows the relationship between Molecular Weight and LogP. The overlap between active and inactive compounds suggests that additional molecular descriptors may be required for accurate prediction.

